

Power-Dense Integrated Motor Drive Using a WBG-Enabled Current-Source Inverter for EV Traction Applications

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Abstract—This article explores the major design considerations associated with the development of a high-power-density integrated motor drive (IMD) utilizing wide bandgap (WBG) power switches in a current-source inverter (CSI) for electric vehicle (EV) traction applications. The rationale for adopting the CSI for this application is presented, followed by a discussion of key design issues associated with all major IMD components, including the CSI itself, dc/dc converter, passive components, control algorithms, CSI-based electric motor, and the cooling system. By focusing attention on the IMD physical integration details, the optimized power electronics and machine can achieve active power densities of 115 kW/L and 55.6 kW/L, respectively. The performance of the proposed IMD system under different operating conditions is carefully evaluated using a combination of simulation and experimental test results. These results provide insights into the advantages that high-power-density IMDs based on CSI and WBG devices can provide for future EV traction drive systems, as well as the remaining technical challenges.

Index Terms—Current-source inverter (CSI), electric vehicle (EV), integrated cooling, integrated motor drive (IMD), passive components, surface inset permanent magnet (SIPM) machine, wide bandgap (WBG) devices.

I. INTRODUCTION

DUE to the global climate crisis caused by escalating greenhouse gas emissions, electric vehicles (EVs) have been undergoing vigorous development worldwide during the past 20 years [1], [2]. To meet the high-performance demands of modern transportation, advanced motor drive systems are increasingly required that achieve high power density by operating at high rotor speeds. Recognizing these imperatives, U.S. Department of Energy (DOE) has set ambitious goals, targeting electric traction drive systems in 2025 EVs to achieve high power densities alongside peak speeds of 20 000 r/min [2]. To address these challenges and

achieve demanding performance milestones, integrated motor drives (IMDs) emerge as a compelling approach that integrates the motor and power electronics into the same physical structure [3], [4]. By eliminating the need for separate housings and bulky high-voltage cables and terminals to connect the inverter to the motor, the total volume and cost of the motor drive system can be significantly reduced [3], [4], [5], [6]. As a result, the IMD configuration is an appealing approach for minimizing the volume and cost of future traction motor drive systems.

Wide bandgap (WBG) semiconductor devices made from silicon carbide (SiC) and gallium nitride (GaN) are appealing for future EV traction drive systems because of their superior performance characteristics, including higher switching speeds, lower ON-state resistances, and higher junction temperatures compared to conventional silicon (Si) devices [7], [8]. The two-level voltage source inverter (VSI) shown in Fig. 1(a) is the dominant choice for EV traction drives that are currently in production. However, there are some limitations for VSI when Si-based switches are directly replaced by WBG switches, such as high dv/dt values in their output waveforms that result in high electromagnetic interference (EMI), serious motor terminal overvoltage, and bearing current risks [9], [10].

The emergence of WBG devices opens opportunities for the long-overlooked current-source inverter (CSI) shown in Fig. 1(b) to provide a promising alternative drive configuration for EV propulsion systems. Recent research has shown that the availability of new WBG power switches makes the CSI an appealing candidate for motor drive applications [9], [10], [11], [12], [13], [14], [15]. The VSI's dc-bus capacitor, which is typically implemented using temperature-limited electrolytic or film capacitors ($<125^\circ\text{C}$), is replaced by a high-temperature dc-link inductor (200°C or higher) for CSIs. This inductor is dramatically reduced in mass and volume compared to those in thyristor-based CSIs from the 1960s because of the much higher switching frequencies that can be achieved using WBG switches.

The three small ac capacitors connected to the output terminals of the CSI function naturally act as low-pass filters in combination with the dc-bus inductor, resulting in much more sinusoidal output voltage waveforms [9], [10]. The smooth excitation waveforms significantly reduce the motor's pulswidth modulation (PWM)-induced losses, common-mode

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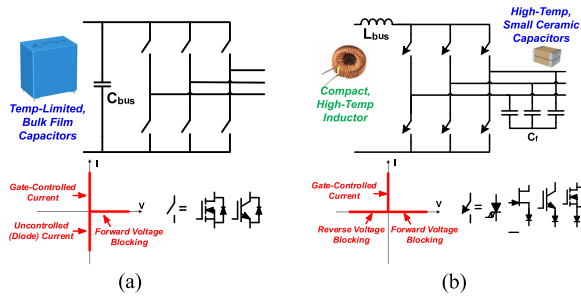


Fig. 1. Schematic of (a) VSI and (b) CSI highlighting their passive components and switch device characteristics.

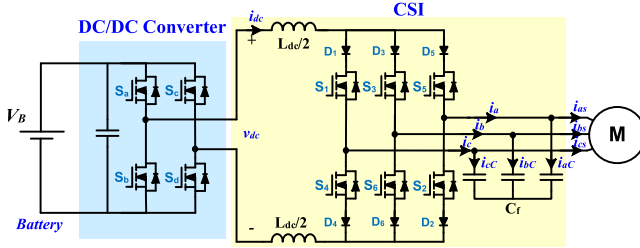


Fig. 2. Configuration of CSI-based motor drive system for electrical vehicle traction application.

EMI, and overvoltage risks at the motor terminals, as well as potential bearing failures caused by dv/dt -induced bearing currents. In addition, the CSI naturally behaves as a boost converter, which can be valuable for extending the machine's constant power speed range [16], [17] and improving the drive system efficiency [18].

A CSI-based traction drive configuration is shown in Fig. 2. The traction drive includes a battery, a bi-directional dc/dc converter, a CSI, and a permanent magnet (PM) machine. The bi-directional dc/dc converter is required to convert the voltage source (battery) into a regulated current source supplying the CSI, and it enables the CSI to charge the battery during machine braking operation [19]. While the presence of this dc/dc converter is not a requirement for a VSI, there are a number of commercial VSI-based EV traction drives that include a dc/dc converter to achieve system advantages by boosting the variable battery voltage to a higher, more stable dc-link voltage [20], [21].

It should also be noted that CSIs require reverse-voltage blocking (RVB) switches for the main inverter switches, typically implemented today using SiC Schottky diodes connected in series with SiC MOSFETs [13]. These extra diodes needed by the CSI increase its component count and conduction losses compared to the VSI. However, these disadvantages will be eliminated when WBG monolithic bidirectional (BD) switches that incorporate the required RVB capability eventually become commercially available [22].

The WBG devices combined with the CSI present an intriguing opportunity for designing power-dense motors that have been optimized for operation with CSI excitation [23]. The surface PM (SPM) machine is known for its high power density, which is appealing for traction applications. However, installation of a carbon fiber containment sleeve to protect SPM machine's magnet from centrifugal forces increases machine cost. An alternative topology that has been

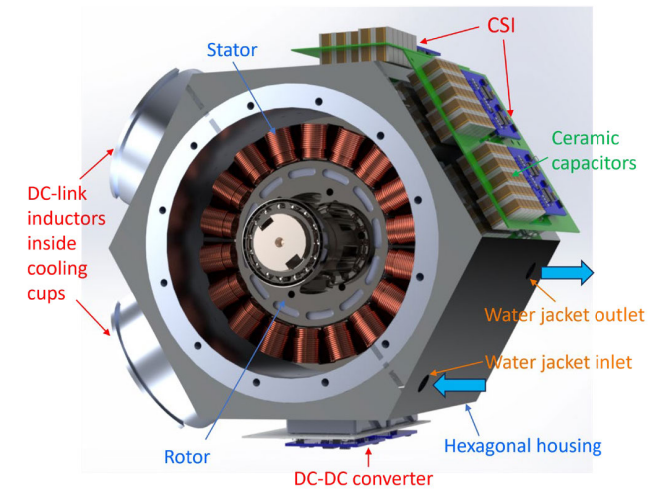


Fig. 3. 3-D rendered drawing of the radially mounted CSI-based IMD.

evaluated for this project is a surface inset PM (SIPM) machine, which uses magnet cavities in the rotor laminations formed by a thin steel web covering the magnets, combined with thin steel bridges connecting the outer web to the inner lamination core. SIPM machines provide the same impressive power density as SPM machines while avoiding the cost of installing a high-tension magnet containment sleeve [24], [25], [26].

Unlike the two separate cooling systems required by conventional traction drives for the motor and power electronics in different housings [27], [28], the proposed IMD incorporates a shared water jacket scheme to cool both the machine and power electronics, increasing the power density and enhancing the overall system efficiency. The stator housing with its integrated water jacket, depicted in Fig. 3, has power electronics components mounted on its outer hexagonal surfaces. This arrangement enables simultaneous cooling of both the motor and the power electronics, optimizing the IMD's thermal management and streamlining the overall design. In addition, forced air ventilation of the end windings and the rotor is incorporated to prevent the magnets from demagnetizing and degrading the IMD's performance.

Wang et al. [29] proposed a 2-kV 1-MW integrated modular motor drive with radially mounted configuration and separate cooling system for power electronics and machine. Compared to the IMD structure in [29], the proposed IMD structure offers several key benefits, including a shared water jacket cooling system for both the power electronics and the machine stator, resulting in a simpler design that significantly reduces the overall volume and cost of the motor drive system. Additionally, using a CSI as the motor drive enables the delivery of sinusoidal voltage and current. Su and Ning [12] applied IGBT-based CSI to traction applications and compared it with VSI, but there is limited published literature addressing the performance, benefits, and challenges associated with applying WBG devices to CSI-based traction motor drive applications. Dai and Jahns [9], Dai et al. [10], Lee et al. [11] proposed a 3 kW WBG-based CSI-IMD system that provides a high-performance design baseline for CSI-based IMDs. However, its axial-mounted power electronics design presents

TABLE I
SUMMARY OF STATE-OF-THE-ART IMD SPECIFICATIONS

Description	Power rating [kW]	Inverter type	Motor type	cooling	VPD [kW/L]	GPD [kW/kg]	Year	Ref.
Multi-phase PM machine with integrated power electronics and fault tolerance	50	6-phase full-bridge	PMSM	air	-	2	2008	[31]
Chevrolet Volt 2nd gen. powertrain where converter and transaxle are integrated	87	2-level VSI	PM assisted SRM	liquid	8.4 (P.E. only)	10.5 (P.E. only)	2015	[32]
Multi-physics design optimization of a GaN-based integrated modular motor drive (IMMD)	8	2-series and 2-parallel VSIs	PMSM	air	0.71	-	2018	[33]
Audi E-tron APA320 electric drivetrain kit	165	2-level VSI	IM	liquid	30 (P.E. only)	20.7 (P.E. only)	2019	[34]
SiC-based five-phase IMMD	110	5-phase VSI	IPM	liquid	-	1.7	2020	[35]
Integrated modular multi-phase drive for Unmanned Aerial Vehicles	1	4 2-level VSIs	AFPM	air	1.43	0.83	2021	[36]
IMD incorporating a WBG-based CSI	3	2-level CSI	SPM	air	9.84	7.31	2021	[11]
Power density boosting techniques for reconfigurable IMMD	-	half-bridge	YASA AFPM	liquid	2.14	-	2022	[37]
Polygon-retrofitted IMMD	3.8	asymmetric H-bridge	SRM	liquid	3.1	-	2022	[38]
Lucid unveils state-of-the-art motorsports electric drive unit	349.5	2-level VSI	IPM	liquid	-	10.96	2023	[39]
2-kV 1-MW 20,000-RPM IMMD for electrified aircraft propulsion	1000	3-level ANPC	SPM	liquid	13.5	9	2023	[29]
GaN-based inverter module for IMMD in all electric aircraft applications	250	series full-bridge	PMSM	air	-	43.96 (P.E. only)	2024	[40]
Proposed IMD using WBG-enabled CSI	100	2-level CSI	SPM	liquid	37.5	-		
Permanent magnet synchronous motor (PMSM) / Surface permanent magnet motor (SPM) / Interior permanent magnet motor (IPM) / Axial flux permanent magnet motor (AFPM) / Yokeless and segmented armature (YASA) / Switched reluctance motor (SRM) / Induction motor (IM)								

challenges for scaling its design up to the much higher power level required for EV traction applications.

The main benefits of the proposed CSI-based IMD system include.

- 1) The power electronics components and machine are integrated into a unified structure, which eliminates the separate housing and bulky high-voltage connecting cables. The total volume and cost of the motor drive system can be significantly reduced, and the CM EMI issue due to the long connection cable can be mitigated.
- 2) The innovative design of the power module-based CSI incorporating ceramic capacitors yields substantial reductions in both overall volume and parasitic inductance within the current commutation loop.
- 3) The sinusoidal output voltage of CSI addresses the high dv/dt problem of VSI and reduces the overvoltage risks at the motor terminals and potential bearing failures caused by dv/dt-induced bearing currents.
- 4) The high-temperature inductor is optimized for achieving high power density, representing strong competitiveness against the VSI dc-bus capacitor.
- 5) The CSI-based SIPM machine without heavy rare-earth magnets is proposed and designed to achieve high power density and high performance.
- 6) The proposed integrated cooling approach is structurally simpler, less expensive, and more effective than traditional separation cooling schemes.

The purpose of this article is to present the design of a power-dense CSI-based IMD system using WBG switches for EV traction applications, including the power electronics, machine, and cooling system. Attention is devoted to the design of each of the major IMD subsystems, including the

dc/dc converter, CSI, SIPM machine, DSP-based controls, and the combined liquid/air thermal management system. Both analytical and measured performance results are provided for the prototype IMD system.

II. PROPOSED POWER-DENSE IMD STRUCTURE WITH INTEGRATED COOLING

Liquid cooling is more effective than air-cooling for achieving high-power density values in demanding applications such as EV traction drives. Radially mounted and axially mounted IMD topologies are two popular alternative configurations for achieving high power density [4]. The axially mounted configurations are preferable in high-torque motors, where the machine cross section tends to be larger [4], [13]. When the power electronics components are axially mounted inside the machine's end bell, the machine's cross-sectional area imposes a significant constraint that can force the axial length of the power electronics to become undesirably long with its own dedicated liquid cooling system [30]. The radially mounted configuration offers the dual advantages of 1) a larger surface area around the machine's circumference to mount the power electronics and 2) a more convenient implementation of a shared heatsink for the machine and power electronics by embedding a liquid cooling jacket in the stator housing. Table I presents a comparison of the key parameters between state-of-the-art IMD technologies and the proposed CSI-IMD, including power rating, inverter topology, motor type, cooling method, volumetric power density (VPD), and gravimetric power density (GPD). It can be observed that the proposed CSI-IMD achieves a higher VPD compared to other IMDs reported in the literature. Based on the DOE roadmap, VPD is a critical design priority for EVs due to strict spatial

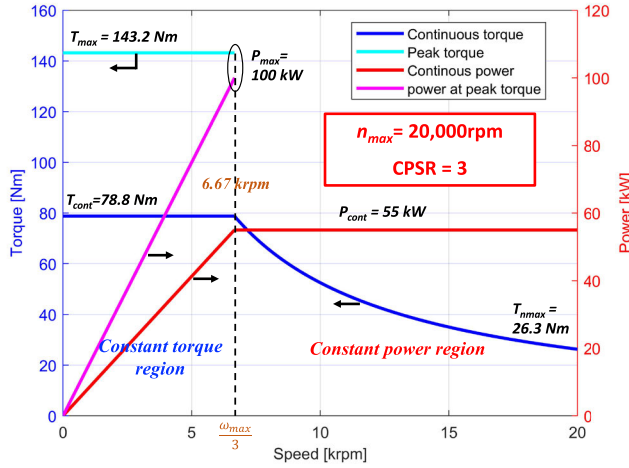


Fig. 4. Specified torque versus speed and power versus speed curves of the CSI-IMD system.

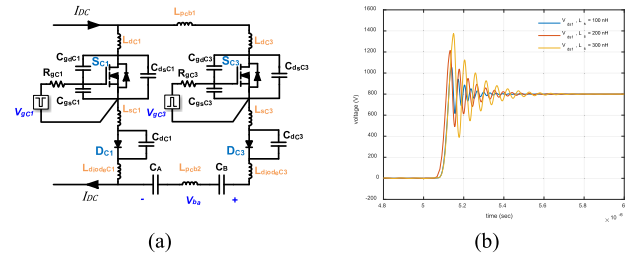
limitations, especially in areas such as “under the hood” or within confined chassis spaces [2]. Thus, the proposed CSI-IMD is focusing on increasing VPD.

Fig. 3 shows the conceptual configuration of the proposed radially mounted CSI-IMD. The power electronics components, including the dc/dc converter, dc-link inductors, and CSI, are mounted on the outer hexagonal surfaces of the machine housing, sharing the same water jacket with the machine stator. The dimensions of the CSI require the power stage to be split between two adjacent hexagonal surfaces, including three-phase legs mounted on these cooled surfaces (two on one surface and one on the other). Printed circuit boards (PCBs) that include the power buses and the output filter capacitors are mounted above the power modules. Next, there are two dc-link inductors, each implementing one-half of the total dc-link inductance $L_{dc}/2$ as shown in Fig. 2. Each of these inductors is a wound toroidal core that is mounted inside an aluminum-cooling cup. These two inductors occupy the next two cooled hexagonal surfaces adjacent to the CSI, as shown in Fig. 3. The dc/dc converter is mounted on the next adjacent hexagonal surface, consisting of two power modules and a PCB with power buses above them.

The specified torque versus speed and power versus speed curves for the CSI-IMD drive system are provided in Fig. 4. The torque curve shows that the rated (continuous) torque (78.8 Nm) and peak torque (143.3 Nm) are required from zero speed up to the corner speed (6667 r/min) where the machine delivers both its rated (55 kW) and peak (100 kW for 30 s) power as shown by the power curve. Constant-power operation at rated power (55 kW) is required at all speeds between the corner speed and the maximum speed of 20 000 r/min. This corresponds to a constant-power speed ratio (CPSR) of 3 ($=20\,000/6667$), which is a typical value for production EV traction motor drives.

III. CSI DESIGN FOR HIGH-POWER DENSITY IMD: CHALLENGES AND SOLUTIONS

There are several technical challenges associated with designing the CSI for this IMD traction application, including the need to provide RVB inverter switches and battery-charging capability, as well as minimizing the size of the CSI



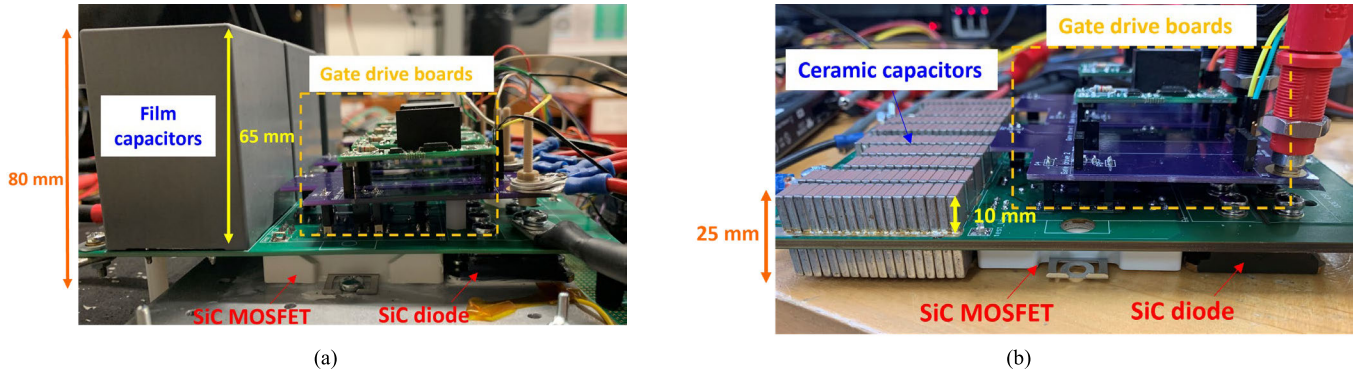


Fig. 6. CSI power stage height comparison of (a) benchtop CSI (Ver. 1) with film capacitors (80 mm) and (b) IMD version CSI (Ver. 2) with ceramic capacitors (25 mm).

in the preceding section, the CSI power modules are split between two hexagonal surfaces with two separate PCBs to make them compatible with the available space on the stator housing. The CSI power stage footprint is 187 mm in axial length and 160 mm in width.

Compared to film capacitors, ceramic capacitors offer multiple performance benefits, including higher power density, higher rms current, lower equivalent series resistance and inductance, and higher temperature operation, which are appealing for the CSI's ac output capacitors [43], [44]. It should be noted that the rms current stress on ac filter capacitors is a crucial design consideration when selecting output capacitors in CSI systems, and ensuring the current capability of the ac capacitor is important [45]. A commercial capacitor unit made by KEMET (0.14 μF , 1200 V, Model CKC33C144KEGLC7805) was selected for its low-loss, low-inductance properties, and its ability to handle extremely high ripple currents without changes in capacitance versus dc voltage and negligible changes in capacitance versus temperature [46]. However, due to the small capacitance of individual ceramic capacitors, a sufficient number of capacitors were paralleled to meet the voltage ripple requirement and the maximum rms current rating for the CSI ac filter. This significantly increased the cost of the ac filter compared to using film capacitors. The IMD version of the CSI (Ver. 2) was designed and fabricated using the selected ceramic capacitors with the capacitance of 7.8 μF for each phase, as shown in Fig. 6(b) [46]. The total volume of the ceramic capacitors for the three-phase ac filter is 0.12 L.

A height comparison of the CSI power boards using film capacitors versus ceramic capacitors is provided in Fig. 6. Ceramic capacitors used in the IMD version of the CSI are significantly more compact compared to the film capacitors used in the benchtop CSI. These ceramic capacitors can also be mountable on the PCB bottom layer as shown in Fig. 6(b), a space that is not utilized in the Ver. 1 CSI. The total CSI power board assembly height was reduced by approximately 70% from 80 to 25 mm by adopting ceramic capacitors. The maximum operating temperature of the ceramic capacitors is 150 $^{\circ}\text{C}$, which is significantly higher than the film capacitors (105 $^{\circ}\text{C}$). The resulting reduction in CSI volume and higher passive component temperature limits are appealing features of IMD power electronics.

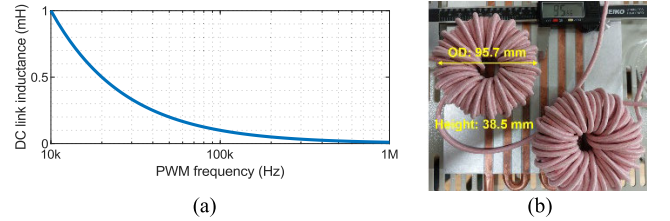


Fig. 7. (a) Inversely proportional relationship between dc-link inductance and PWM frequency. (b) Fabricated toroidal core inductors using litz wire.

TABLE II DC-LINK INDUCTOR PARAMETERS ($L_{dc}/2$ IN EACH BUS)			
Parameters	Value	Parameters	Value
Max. current ripple (A)	25.8	Number of turns	73
Peak l-l voltage, V_{LL} (V)	800	Core type	Powder iron
Inductance, $L_{dc}/2$ (μH)	48.4	Wire type	Litz wire
Rated current (A)	221	Wire size	AWG 8 (equiv.)
PWM freq. (kHz)	80	Active volume	0.1 L/inductor

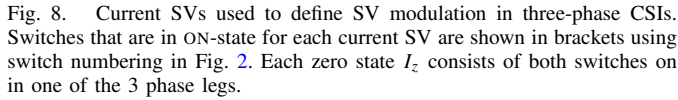
B. Compact DC-Link Inductor Design for IMD

The high switching frequency of the WBG devices is a critical enabler for the power-dense CSI, especially for the passive components. The required dc-link inductance L_{dc} for a desired maximum peak-to-peak current ripple $\Delta i_{dc,pp}$ and PWM frequency f_{PWM} is given by

$$L_{dc} = \frac{V_{LL}}{4 \cdot \Delta i_{dc,pp} \cdot f_{PWM}} \quad (1)$$

where V_{LL} is the peak amplitude of the CSI's line-to-line output voltage [47]. Thus, the dc-link inductance L_{dc} requirement for a given $\Delta i_{dc,pp}$ is inversely proportional to the switching frequency f_{PWM} , as shown in Fig. 7(a). Lowering the required dc-link inductance reduces the number of inductor winding turns as well as the cross-sectional area of the inductor core, shrinking the inductor volume.

By increasing the switching frequency of the CSI to 80 kHz, the inductor size was reduced to a total volume of 0.2 L following a process similar to that presented in [48]. The parameters used to calculate the total link inductance L_{dc} via (1) are provided in Table II, and the resulting total inductance $L_{dc} = 96.8 \mu\text{H}$ was split in two, with half in each dc bus (see Fig. 2) to reduce the conducted common-mode EMI. Key parameters of each inductor $L_{dc}/2$ are also presented



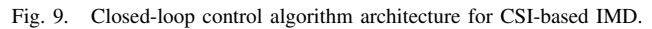
C. Pulsewidth Modulation and Control of CSI

The CSI SV PWM dwell times can be derived to be

where T_n is the dwell time (i.e., duration) of the active current vector I_n , T_{n+1} is the dwell time of the active current vector I_{n+1} , T_z is the dwell time of the zero-current vector I_z , and T_s is the PWM frequency period ($=1/f_{\text{PWM}}$). The reference current vector angle within each sector (denoted as θ in Fig. 8) is given by

where n is the sector number ($=1-6$), and θ' is the absolute angle of t_{abck}^* with respect to the α -axis as shown in Fig. 8.

The controllers are designed in the dq synchronous reference frame, similar to common practice in the VSI case. However, the duality between the CSI and VSI leads to a control configuration with an inner voltage loop and an



Unlike the motor drive with VSI, the CSI traction drive system needs an active front-end power stage to convert the battery voltage into a controlled current source with a value of i_{dc} . The developed traction drive system uses a bi-directional dc/dc buck converter with dc-link current control capability. The dc-link current i_{dc} is measured and fed back to the dc current controller. The dc-link current command i_{dc}^* is typically set to equal the CSI's output current command i_{dqk}^* (i.e., $i_{dqk}^* = |i_{abck}^*|$) in order to minimize the CSI's conduction losses by operating at unity modulation index (i.e., $m = 1$), as shown in Fig. 9 [50]. A PI controller is used for the dc current controller. The output of the dc current controller is the dc/dc converter's output voltage command v_{dc}^* , which is synthesized using PWM.

The front-end converter ensures that the CSI is capable of regenerative operation as both a motor and regenerative brake [19], [46], a critical feature for vehicle traction drives. Unlike the VSI, no reversal of the dc link current direction is needed to reverse the power flow and charge the battery.

The dc/dc converter consists of two SiC half-bridge modules forming an H-bridge topology (Fig. 10) that operates in four switching states, as shown in Fig. 11. Same as the CSI, the dc/dc converter can deliver 55 kW rated power and 100 kW peak power for 30 s. Fig. 11(a) and (b) shows the active motoring and battery charging states, respectively, of the

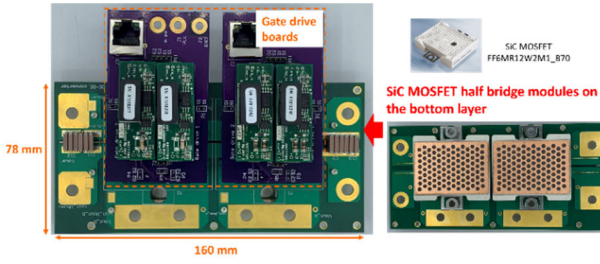


Fig. 10. DC/DC converter implementation using 2 SiC half-bridge modules.

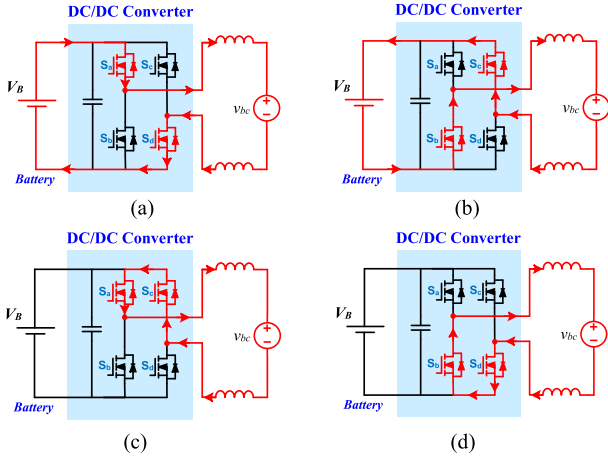


Fig. 11. DC/DC converter operating states. (a) State I. (b) State II. (c) State III. (d) State IV.

CSI-based motor drive system, where the simplified CSI-plus-motor model consists of a dc voltage source with two series inductors [19]. The freewheeling states are achieved by concurrently turning on either the two top switches or the two bottom switches, as shown in Fig. 11(c) and (d), respectively. It should be noted that the dc/dc output voltage v_{dc} in Fig. 11 is bipolar, varying between $\pm V_B$, where V_B is the battery (i.e., dc source) voltage. For motoring operation, the dc/dc converter alternates between states I and III or states I and IV to regulate the dc-link current of CSI. For regenerative braking operation, the dc/dc converter alternates between states II and III or states II and IV to regulate the battery charging current.

It should be noted that switches S_b and S_c only conduct current in the reverse (i.e., source-to-drain) direction, representing diode operation. Actively gating these two devices when they are conducting reverse current corresponds to well-known synchronous rectifier operation [51], reducing the total losses in these two switches compared to the alternative of using the internal diodes alone without active gating.

Although the front-end buck converter can be implemented using only two active switches and two diodes [19], [52], the front-end dc/dc converter built using two SiC MOSFET half-bridge modules (FF6MR12W2M1_B70) as shown in Fig. 10 offers two advantages: 1) today's SiC half-bridge module is smaller than a two-package (discrete MOSFET + discrete diode) implementation and 2) the conduction loss of an ON-state SiC MOSFET is lower than that of a SiC Schottky diode, increasing the converter's efficiency.

IV. ELECTRIC MACHINE DESIGN FOR POWER-DENSE IMD

A. WBG-Based CSI Impact on Electrical Machine Design

The higher switching frequency of the WBG-enabled CSI allows the electric machine to operate at higher speeds with higher pole numbers, which help to reduce the machine mass and volume. In addition, higher switching frequencies help reduce the amplitude of harmonic components in the CSI's ac output waveforms, which are beneficial for reducing losses in both the stator core laminations and windings [18].

While conventional VSIs operate as voltage buck-type power converters with output voltages that cannot exceed the dc link voltage, the CSI output voltage can be higher than its dc link voltage because it operates as a voltage boost-type inverter. The steady-state relationship between the CSI output voltage and the dc-link voltage as a function of the modulation index m is expressed as [17]

$$V_{pk(CSI)} = \frac{0.67}{m \cos(\theta + \varphi)} V_{dc} \quad (4)$$

where $V_{pk(CSI)}$ is the peak line-to-neutral fundamental voltage of the CSI, m is the CSI modulation index defined previously, θ is the angle between the motor phase voltage and phase current (i.e., power factor angle), and φ is the angle between the CSI output current and motor phase current [17]. It can be observed that the CSI's output voltage increases inversely with the modulation index m , making it possible to increase the CSI's output voltage much higher than V_{dc} by lowering the modulation index below 1. In contrast, the peak value of V_{dc} for the VSI is $0.955 V_{dc}$.

The ability to deliver higher output voltage from the same battery is beneficial for traction drive applications because of machine performance limits imposed on all ac machines when the maximum terminal voltage is limited by the inverter at high speeds. More specifically, the fixed maximum voltage available from VSIs makes it necessary to transition into flux weakening constant-power operation at speeds above a corner point speed determined by the machine's design [49]. Achieving wide speed ranges of constant-power operation in VSI-excited PM machines requires careful machine design tradeoffs that can penalize the machine's power density, efficiency, and power factor. Adopting a CSI that is not subject to the VSI's maximum voltage limit opens the door to designing new PM machines that can exceed the performance of VSI-optimized machines because of the extra degree of freedom provided by voltage-boosting. For a CSI motor drive, the maximum voltage that can be applied to the machine is limited by the maximum voltage limits of the inverter switch devices instead of the battery voltage, opening opportunities to improve the machine's performance metrics [17].

Another benefit of the voltage-boost capability is that CSI-based machines can be designed for traction applications with lower inductance compared to VSI-based machines. This can be valuable because some types of PM machine designs, such as axial-airgap machines or machines with airgap windings (i.e., toothless), are often associated with low-phase inductances. In addition to the limitations on wide constant-power speed ranges, low inductance often results in unacceptably high current and torque ripple with VSI excitation because of the high harmonic content of the PWM

TABLE III
DESIGN PARAMETERS OF SIPM MACHINE

Magnet materials	Arnold N50	Stator OD [mm]	167.3
Lamination materials	Arnon 5	Stator ID [mm]	107.6
Number of stator slots	18	Rotor OD [mm]	105.8
Number of rotor poles	12	Rotor ID [mm]	35
Parallel circuit	6	Stack length [mm]	100.7
Corner speed [rpm]	6,667	Magnet thickness [mm]	8.88
Max speed [rpm]	20,000	Running airgap [mm]	0.9
Maximum fundamental frequency [Hz]	2,000	Number of turns per coil	48
Rated Power [kW]	55	Peak Power [kW]	100
Rated current @ 55 kW [A_{rms}]	81	Peak current @ 55 kW [A_{rms}]	156
Rated line-line voltage @ 55 kW [V_{rms}]	444	Peak line-line voltage @ 55 kW [V_{rms}]	622

voltage waveforms. In contrast, the output capacitors of CSI act as effective low-pass filters in combination with the dc-link inductors for suppressing these PWM switching harmonics. The resulting voltage and current waveforms are highly sinusoidal, sharply reducing the current/torque ripple effects.

B. SIPM Machine Design for IMD

Based on DOE Roadmap 2025 performance targets [2], a surface-inset SIPM machine was designed using a fractional-slot concentrated winding (FSCW) to deliver 55 kW continuous power and 100 kW peak power while achieving a peak active power density of 50 kW/L without using heavy rare-earth material. The design parameters of the SIPM machine are shown in Table III. More specifically, Arnold N50 neodymium-iron-boron (NdFeB) magnets without any dysprosium were selected. The lamination steel is Arnon 5, a high-quality Si steel, that is, 5 mils thick. A high maximum speed (20 krpm) was chosen to increase the machine's power density. A FSCW configuration with concentrated windings was selected to help boost the phase inductance values [53] as well as to reduce the axial length of the end-turn windings.

Fig. 12 shows the predicted flux density distribution of the SIPM machine using finite element analysis (FEAs) when delivering 55 kW at 6667 r/min based on JMAG simulation. At this rated power operating point, magnetic saturation on the stator core is mild with a peak flux density of approx. 1.3 T. Fig. 13 shows the predicted output power and torque versus speed envelopes, indicating that the machine can deliver rated power over a 3:1 speed range from 6667 to 20 000 r/min. The reluctance torque made possible by the modest magnetic saliency of the SIPM rotor design helps to ensure that sufficient torque is produced in the constant-power speed region.

The proposed SIPM machine has been successfully fabricated. Overlaid measured and FEA-predicted back EMF waveforms are provided in Fig. 14. The waveshapes and amplitudes of the measured and predicted back EMF waveforms match closely. More specifically, the amplitude of the fundamental component of the measured back EMF waveform is 2.3% lower than the FEA-predicted value.

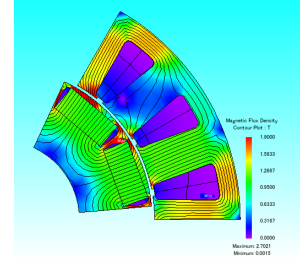


Fig. 12. FEA-predicted flux density distribution for 55 kW, 6667 r/min operation of SIPM machine.

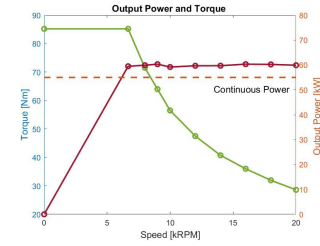


Fig. 13. FEA-predicted torque and power versus speed operating envelopes for SIPM machine.

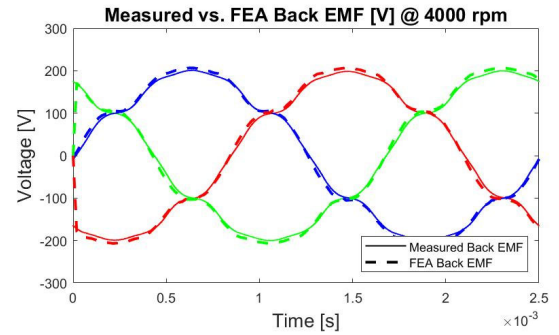


Fig. 14. Overlaid FEA-predicted and measured three-phase back EMF waveforms of SIPM machine @ 4000 r/min at room temperature, showing a close match with a 2.3% reduction in the measured fundamental component amplitude compared to the FEA-predicted value.

V. INTEGRATED COOLING SYSTEM DESIGN OF IMD

For a high-performance IMD for EV applications, heat losses generated by both the power electronics and the motor must be dissipated via a well-designed cooling system to maintain all component temperatures within their maximum limits. The traditional strategy for conventional motor drives with segregated motor and drive units is to use an independent cooling system for the motor and drive, adding undesired volume and mass to the drive system. IMDs provide valuable opportunities to increase the drive's power density by adopting an integrated cooling system. The integrated cooling system designed for this IMD uses both a water jacket for the stator core and power electronics cooling, and forced-air convection cooling for the rotor and stator end windings.

A. Power Loss Distribution of IMD System

The IMD power losses were estimated using a combination of PLECS simulations and FEA. These machine and power electronics losses represent the heat generated inside each of the major IMD components at different operating points, and those loss values were subsequently used as inputs to

TABLE IV
PREDICTED POWER LOSSES OF KEY IMD COMPONENTS

	Losses	55 kW 6,667 rpm	55 kW 20,000 rpm	100 kW 6,667 rpm
Motor	Winding [W]	1345	1956	5064
	Iron loss [W]	325	476	453
	Magnets [W]	19	188	64
	Windage [W]	2.5	67	2.5
	Total motor loss [W]	1691	2687	5583
CSI (80 kHz)	CSI MOSFET [W]	306	626	911
	CSI diode [W]	270	421	604
	DC/DC converter [W]	192	363	604
	DC-link inductor [W]	313	652	1148
	Total P. E. loss [W]	1081	2062	3267
IMD	Total IMD loss [W]	2772	4749	8850
	IMD efficiency [%]	95.2	92.1	91.9

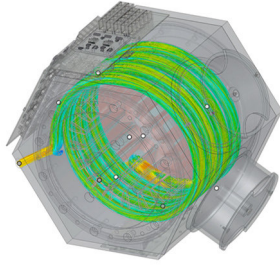


Fig. 15. Shared water jacket for machine and IMD power electronics.

the thermal simulation as the heat sources. Table IV shows the predicted losses for three key operating points along the envelopes in Fig. 4 that were used in the thermal analysis.

B. Shared Water Jacket Design

The cooling system integrates a shared water jacket into the stator housing to efficiently dissipate heat from both the stator assembly and the power electronics, as shown in Fig. 15. This dual-sided cooling jacket features hexagonal outer surfaces to conveniently mount the power electronics components, and a circular inner bore to accommodate the cylindrical stator core. The objective of this design is to achieve effective heat transfer from all components that come into direct contact with the inner and outer surfaces of the stator housing, thereby playing a key role in the IMD's thermal management. The internal coolant channel follows a spiral path between the two axial ends of the housing in order to minimize the pressure drop and maximize cooling efficiency.

The coolant chosen for this water jacket is distilled water, selected for its compatibility with the lab chiller. To ensure sufficient heat dissipation performance, a minimum flow rate of 2 L/min was chosen based on the correlation between the flow rate and the equivalent channel wall heat transfer coefficient. This flow rate corresponds to the transition from laminar to turbulent flow, typically occurring when the Reynolds number exceeds 2300. In practice, the Reynolds number should exceed 10 000 to guarantee fully turbulent flow for optimum heat transfer. This would require a flow rate of approximately 8.5 L/min to achieve maximum heat removal and optimal cooling system performance.

Ensuring adequate cooling for all power electronics components that share the water jacket is a critical requirement

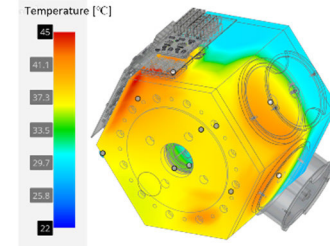


Fig. 16. Predicted outer surface temperature of water jacket for 55 kW at 6667 r/min (rated power).

TABLE V
PREDICTED PEAK TEMPERATURES OF POWER MODULE DEVICE
JUNCTIONS, INDUCTORS, AND MACHINE: COOLANT
FLOW = 2 GPM (7.56 L/MIN); INLET TEMP. = 20 °C

Components	S.S. 55 kW 6,667 rpm [°C]	S.S. 55 kW 20,000 rpm [°C]	30s 100 kW 6,667 rpm (From S.S) [°C]
CSI MOSFETs (case)	39.6	49.0	50.7
CSI MOSFETs (junction)	66.7	84.0	86.0
CSI diodes (case)	34.1	54.9	59.6
CSI diodes (junction)	50.1	65.2	69.6
DC/DC MOSFETs (case)	38.8	48.1	44.4
DC/DC MOSFETs (junction)	70.6	92.6	97.1
Inductors (cooling cup)	38.8	48.1	44.4
Inductors (winding)	58.8	68.8	66.4
Machine winding	112.5	124.4	146.4
Rotor magnets	52.4	72.8	53.5

of this design. Fig. 16 provides the predicted outer surface temperature distribution of the housing at rated power (55 kW) and corner speed (6667 r/min) operation, with a coolant flow rate of 6 L/min and an inlet temperature of 20 °C. The predicted temperatures based on thermal FEA simulation and analytical prediction for the power module devices (junction and case), inductors (winding and aluminum cooling cup), and machine windings at key operation points are presented in Table V. It should be noted that the junction temperatures of the semiconductor devices and the inductor winding temperatures are estimated by calculating the temperature rise based on power losses and the thermal resistances of the power devices and thermal interface materials.

Since the maximum specified junction temperature for the SiC MOSFETs and diodes are 150 °C and 175 °C, respectively, the predicted junction temperatures for all three of these operating points are well within their limits. Similarly, the predicted maximum temperatures of the inductors and stator windings are within their maximum limits as well. These results build confidence in the effectiveness of the water jacket cooling system for managing the operating temperatures of all key machine and power electronics components over the IMD's complete operating envelope.

C. Forced Air Cooling Design

Since the selected rotor magnets do not contain any heavy rare earth material to raise their safe operating temperature,

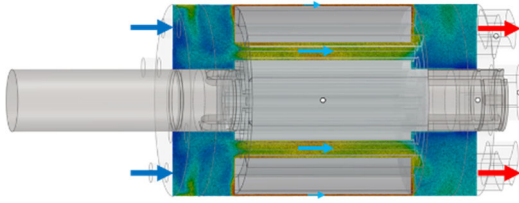


Fig. 17. Forced airflow paths inside the machine.

they must be prevented from exceeding 80 °C during the IMD's operation. As shown in Fig. 17, the forced-air convection cooling system is designed to inject pressurized air into the end chamber at the machine's drive end so that it can first cool the stator end windings in that chamber. It then flows axially through six ducts in the rotor hub as well as through the rotor/stator airgap. Finally, the air is released into the atmosphere through ports in the endplate at the nondrive end of the machine after cooling the end windings in that end chamber.

The designated airflow rate is 0.01 m³/s, based on the capability of the lab facility air. Simulation results from computational fluid dynamics (CFDs) analysis have predicted peak magnet temperatures for rated power operation of 52.4 °C for 6667 r/min and 72.8 °C for 20 000 r/min. These values indicate that the forced air-cooling system should be capable of limiting the maximum rotor magnet temperature within safe limits for all key operating conditions. However, the thermal margins are significantly smaller than for the power electronics components cooled by the water jacket, and the assumption about the ambient air temperature has a major impact on the results. These results highlight the fact that the elimination of heavy rare earth from the NdFeB magnets significantly increases the challenges for the rotor cooling system design.

VI. DEVELOPMENT AND TESTING OF IMD

A. Development of Power-Dense IMD System

The aluminum inner and outer stator housing assemblies that were fabricated for the prototype IMD unit are shown in Fig. 18. The two-machined cylindrical housing components were assembled with a press (interference) fit to form the spiral-cooling channel. Circular O-rings inserted into the two machine endplates are used to achieve water-tight seals.

The final assembled IMD with all power electronic components mounted on the hexagonal surfaces of the outer stator housing is shown in Fig. 19. Low-impedance copper bars and screws are utilized to connect the two separate CSI power stages to minimize the parasitic inductance between the CSI phase legs. Shielded Ethernet cables and connectors are employed to connect the sensor board and gate drive boards to the control board, effectively reducing the risk of EMI. The active volumes of the power electronics and electric machine components that were estimated using the mass properties tool in SolidWorks are presented in Table VI. The active power density values of the machine and power electronics both meet the DOE targets [2]. A more detailed comparison of power density between CSI- and VSI-based motor drive systems can be found in the previous work [46].



Fig. 18. Machined inner (top) and outer (bottom) housing for the IMD, with an integrated water jacket in the inner housing.

TABLE VI
CALCULATED ACTIVE VOLUME AND POWER DENSITY OF POWER ELECTRONICS AND MACHINE ASSEMBLIES

Components	Active Volume [L]	Components	Total Volume [L]
Stator core	0.92	SiC MOSFETs	0.15
Stator winding	0.19	SiC diodes	0.07
Rotor core	0.40	DC-link inductors	0.2
Magnets	0.27	PCB boards	0.33
Sleeve	0.02	Ceramic capacitors	0.12
Machine total	1.80	P.E. total	0.87
Machine active power density	55.6 kW/L	Power electronics active power density	115 kW/L
IMD active power density		37.5 kW/L	

B. Experiment Results

The performance of the CSI-based IMD system is experimentally validated under different operating conditions using a high-torque dynamometer. Given the capabilities of the laboratory test setup, the IMD system can be tested up to 65 Nm (more than 80% of the rated torque) and 2000 r/min. The PWM switching frequency of the CSI and dc/dc converter is set to 80 kHz in all operating conditions. The measured waveforms under 3 torque and speed operating points (Condition #1: 50% rated torque, 1000 r/min rotation speed, 60 A dc-link current; Condition #2: 80% rated torque, 1000 r/min rotation speed, 91 A dc-link current; Condition #3: 80% rated torque, 2000 r/min rotation speed, 91 A dc-link current) are provided in Fig. 20. It can be observed that the CSI delivers output voltage and current waveforms that are much more sinusoidal compared to those of a conventional VSI.

The FFT analysis of the voltage and current waveforms under these three operating conditions is shown in Fig. 21. It can be observed that the THD of the current waveforms ranges from 1.35% to 2.53%, while the THD of the voltage waveforms is significantly higher, ranging from 13.55% to 17.53%. The higher THD of the voltage waveforms is caused by the switching oscillation between the parasitic inductance in the current commutation loop and the output capacitors during

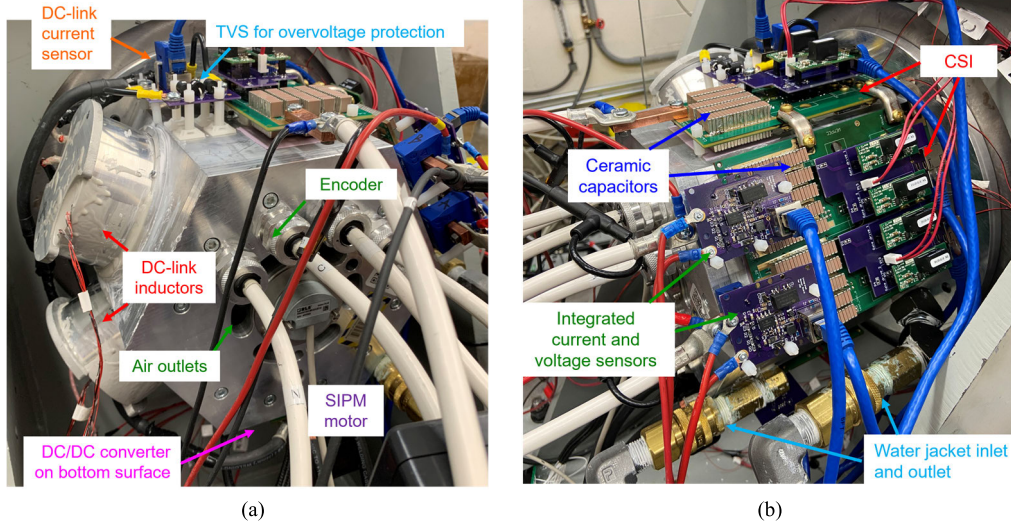


Fig. 19. Two views of the fabricated prototype IMD unit highlighting CSI components mounted on stator housing. (a) Left-side view. (b) Right-side view.

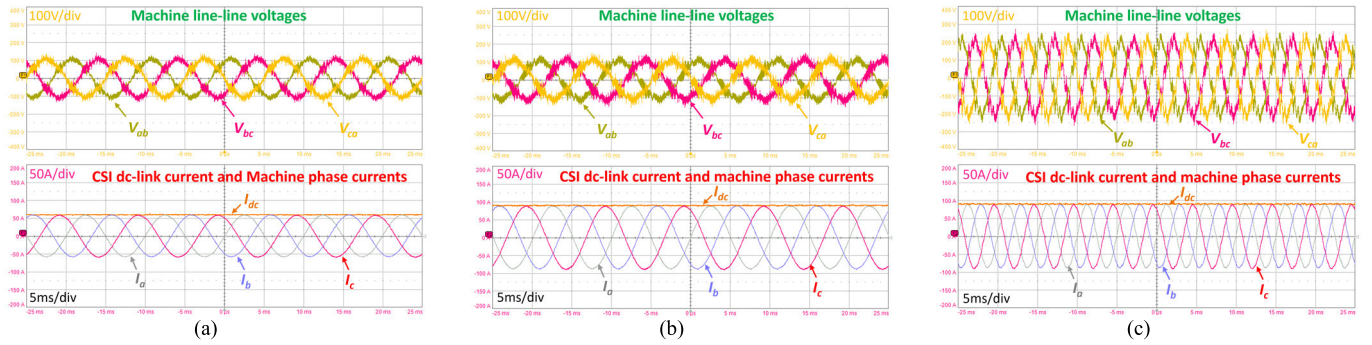


Fig. 20. Experimentally measured waveforms of machine line-to-line voltage and phase current, and CSI dc-link current under different torque and speed conditions. (a) Operating condition #1: 41.3 Nm ($\sim 50\%$ rated torque) and 1000 r/min, dc-link current is 60 A. (b) Operating condition #2: 65.2 Nm ($\sim 80\%$ rated torque) and 1000 r/min, dc-link current is 91 A. (c) Operating condition #3: 63.2 Nm ($\sim 80\%$ rated torque) and 2000 r/min, dc-link current is 91 A.

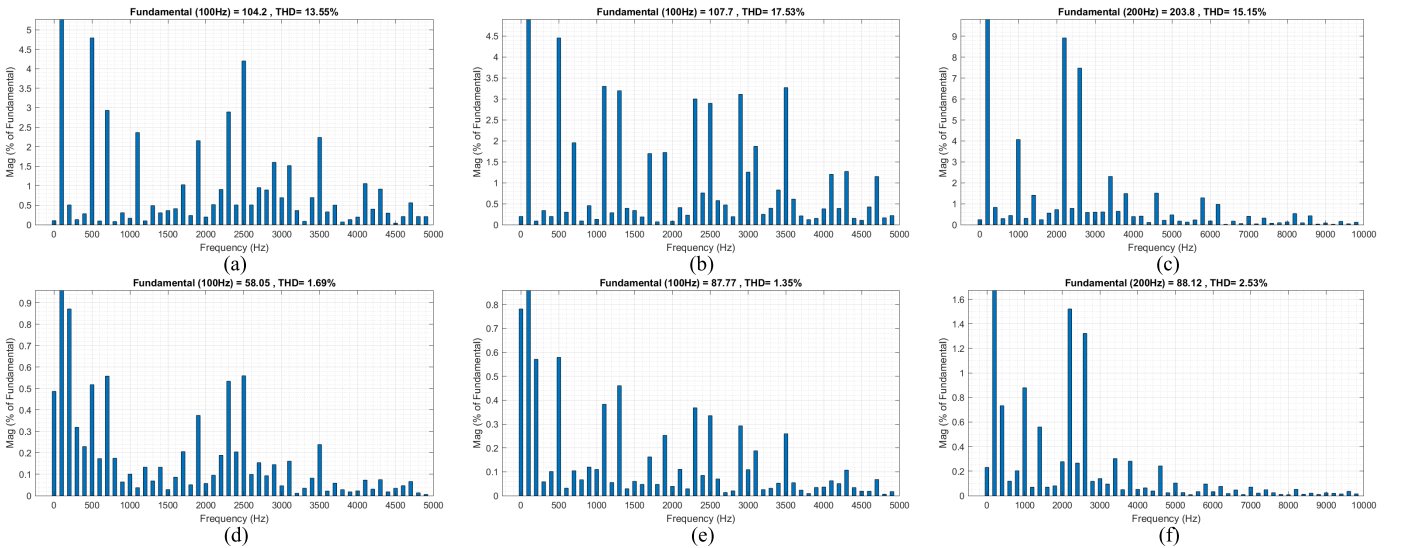


Fig. 21. FFT analysis of CSI output voltage and current waveforms. (a) Voltage waveforms FFT for Condition #1. (b) Voltage waveforms FFT for Condition #2. (c) Voltage waveforms FFT for Condition #3. (d) Current waveforms FFT for Condition #1. (e) Current waveforms FFT for Condition #2. (f) Current waveforms FFT for Condition #3.

switching events, as mentioned in Section III. Additionally, it should be noted that the back EMF of the SIPM machine is not purely sinusoidal, exhibiting significant fifth- and seventh-

order harmonics, as shown in Fig. 14. These harmonics increasingly disturb the machine operation, causing a growing impact on the voltage waveforms and, to a lesser degree, on the

TABLE VII
MEASURED STEADY-STATE TEMPERATURES OF POWER MODULE
DEVICES, INDUCTORS, AND MACHINE: COOLANT
FLOW = 6 L/MIN; INLET TEMP. = 20 °C

Components	91A dc-link current, 80% torque, 1,000 rpm [°C]	91A dc-link current, 80% torque, 2,000 rpm [°C]
Measured CSI MOSFET case	36.9	38.1
Calculated CSI MOSFET junction	52.5	56.1
Measured CSI diode case	33.9	34.2
Calculated CSI diode junction	47.6	49.2
Measured DC/DC MOSFETs case	42.8	46.0
Calculated DC/DC MOSFETs junction	61.1	69.4
Inductors	48.8	52.4
Machine Winding	68.0	71.3

current waveforms, making them less sinusoidal with some ripple components.

To effectively monitor the thermal dynamics of the prototype IMD, 13 resistive temperature detectors (RTDs) have been strategically placed inside the machine and inductor cooling cup to measure the temperatures of the machine windings and inductors, and on the bottom surfaces of the power electronics components to capture their case temperatures. The measured temperatures for the power module devices, inductors, and machine laminations are presented in Table VII. It should be noted that the RTDs measure the case temperatures of SiC MOSFETs and diodes for the CSI and dc/dc converter, while the junction temperatures are estimated based on power losses and thermal resistances of the power devices. The measured results verified that the cooling system is sufficient to some extent.

VII. CONCLUSION

This article has presented a new WBG-enabled IMD incorporating a CSI that achieves high power density for EV propulsion drive applications. The advantages offered by CSIs for EV traction drives in IMD configurations have been discussed, as well as the critical role played by WBG switches in shrinking the size of the CSI's passive components.

The design details of a prototype version of this CSI-based IMD system have been examined, including the power electronics, machine, controls, and cooling system. A thorough analysis of the CSI-IMD drive system has been carried out, providing the basis for performance predictions and the selection of key passive component values. The results of this analysis have been used to finalize the designs of the high-power-density CSI and its associated dc/dc converter. All components have been tailored for compatibility with the chosen IMD topology that mounts the power electronics around the outer circumference of the machine.

The impact of the WBG-based CSI on the design of a high-performance PM machine has been investigated. This work resulted in the design of a power-dense surface-inset PM machine that was optimized for CSI excitation while avoiding the need for heavy rare-earth elements in the NdFeB magnets.

The design of an integrated cooling system has also been presented that features a shared water jacket that cools both the machine and power electronics. This is complemented by forced air ventilation of the stator end windings and rotor, ensuring effective cooling of the entire IMD drive system.

Finally, details of a prototype version of the WBG-based CSI-IMD system have been presented that highlight key features of the fabricated unit. The resulting active volumetric power densities of the power electronics and machine are 115 and 55.6 kW/L, exceeding the demanding DOE target values. Measured performance results from the prototype drive system have been presented that confirm the promising advantages of the CSI-IMD configuration for future electrified propulsion drive applications.

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